

“Green UPS are designed not to pollute the environment”



“Your life changes the moment you make a new, congruent and committed decision”—this statement goes well with the very dynamic and diligent R.K. Bansal, founder and managing director of Uniline Energy Systems.

Incorporated in 1989, today Uniline is the third largest Indian manufacturer of UPS systems and one of the fastest growing UPS companies in India in IGBT technology. The company designs and manufactures high-quality UPS systems, inverters and other power-conditioning products and provides power-conditioning solutions to all the industry segments, viz, PSUs, banking, ministries, medical institutions, telecom and SMEs.

Uniline MD speaks to Uma Bansal, deputy editor at EFY, about UPS systems and related issues

■ **R.K. BANSAL**
FOUNDER AND MANAGING DIRECTOR OF UNILINE ENERGY SYSTEMS

What's the role of IGBT in a UPS system? What's next after IGBTs?

Earlier, UPS were manufactured with a simple transistor that acted like a switch. The switching frequency was very low. Then came MOSFETs, which had a very high switching frequency resulting in high efficiency and low noise of the system. However, the online resistance of MOSFETs was very high, requiring use of more MOSFETs for the same power output and dissipation mechanism for the heat generated.

IGBTs have advantages of both transistors and MOSFETs. These offer the advantages of high switching frequency and low online resistance minus disadvantages of MOSFETs and transistors.

IGBTs themselves are going through a lot of changes. Currently, the IGBT being used is sixth-generation. Earlier, IGBT rating and DC voltage were low. Now IGBTs are available up to 600A and 1200V. Also, two IGBTs are being packaged into a single chip.

Talking of the technology next to IGBT, intelligent power modules have just started appearing. These are IGBTs with a complete electronic circuit that takes care of output voltage, output frequency and driver.

What are the techniques being used to maximise the efficiency of UPS systems?

The efficiency depends on two-three factors. The more the switching frequency, the better the efficiency. The more the DC voltage closer to the output, the better the efficiency. The type of transformer core and wires also affect the switching frequency.

What's a multifunction UPS system?

There are many applications in a plant that can't wait for power. A multifunctional UPS system can provide backup for almost all the loads, not just computers, used in a plant. So there is no need for the generator to start. This is particularly useful for medical electronics.

How green are the Green UPS?

Green UPS are designed not to pollute the environment. For IGBT-based rectifiers, the input power factor is 0.99. The earlier thyristor-based SCR's had maximum power factor of 0.8 only. Also, these had up to 25 per cent input current harmonics, which distorted the line. With IGBT, the input current harmonics are less than 3 per cent and the input current drawn from the electricity grid is